

SS3 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

FIRST TERM

WEEK 1: REVIEW OF SS2 WORK

TOPIC: Q&A, RECAP

Introduction

This week serves as a crucial foundation for the advanced topics in SS3. We will refresh our knowledge of all fundamental concepts from SS2, ensuring everyone has a solid grasp of basic diagnostics, component identification, and essential repair techniques. This is a highly interactive week, so come prepared with your questions.

A. Core computer hardware concepts (recap)

1. COMPONENT IDENTIFICATION:

- **Motherboard:** The main circuit board. Key parts: CPU socket, RAM slots, PCIe slots, SATA ports, I/O panel (USB, audio, LAN), chipset, BIOS/UEFI chip.
- **CPU (Central Processing Unit):** The "brain." Understand models (Intel Core i-series, AMD Ryzen) and socket types (LGA, PGA).
- **RAM (Random Access Memory):** Volatile memory. Types: DDR3, DDR4, DDR5. Form factors: DIMM (desktop) vs. SO-DIMM (laptop).
- **Storage:** HDD (Hard Disk Drive - magnetic, spinning platters) vs. SSD (Solid State Drive - flash memory, faster). Interface: SATA vs. NVMe (M.2).
- **PSU (Power Supply Unit):** Converts AC to DC power for all components. Understand wattage and connectors (24-pin, 8-pin CPU, PCIe 6+2 pin).
- **Peripherals:** Input (keyboard, mouse), Output (monitor, printer), Storage (external HDD).

2. BASIC DIAGNOSTICS & TROUBLESHOOTING:

- **POST (Power-On Self-Test):** The initial hardware check. Beep codes and on-screen messages.

- **Common Problems:**
 - **No Power:** Check wall outlet, PSU switch, PSU itself.
 - **No POST / No Display:** Check RAM (reseat), graphics card, monitor connections.
 - **OS Not Booting:** Check storage drive connection, BIOS boot order, look for "No Bootable Device" errors.

B. Core GSM (phone) repair concepts (recap)

1. COMPONENT IDENTIFICATION (SMARTPHONE):

- **Main Board (Logic Board):** Contains the SoC (System on a Chip), RAM, storage (eMMC/UFS), and power management ICs (PMIC).
- **SoC:** Integrates the CPU, GPU, and modem (network).
- **Battery:** Lithium-ion (Li-ion). Understanding voltage and capacity (mAh).
- **Daughterboard:** A smaller board, often containing the charging port, microphone, and network antenna connections.
- **Flex Cables:** Flexible flat cables connecting the main board to the screen, daughterboard, cameras, and buttons.

2. BASIC REPAIR TECHNIQUES:

- **Tool Usage:** Proper use of the precision screwdriver set, prying tools (spudgers, opening picks), and heat gun (for removing screens/back glass).
- **ESD (Electrostatic Discharge) Safety:** The critical importance of using an anti-static mat and wrist strap.
- **Common Repairs:** Screen replacement, battery replacement, charging port replacement (daughterboard swap).
- **Multimeter Basics:** Checking for continuity (broken traces/cables) and voltage (battery, charging port output).

C. Q&A session

- This part of the lesson is driven by students. Topics to clarify:
 - "What is the difference between BIOS and UEFI?"

- "How do I *really* test a PSU?"
- "What is the safest way to open a phone with a glass back?"

Evaluation questions

1. What is the function of the POST?
2. What is the main difference between an HDD and an SSD?
3. Name three key components found on a smartphone's logic board.
4. What does "ESD" stand for, and why is it a major concern?
5. What tool would you use to check for a broken wire (continuity)?

Assignment questions

1. Draw a simple block diagram of a desktop computer, showing how the CPU, RAM, Storage, and GPU connect to the motherboard.
2. A laptop user complains their device is "too slow." List three hardware and two software issues that could be the cause.
3. Describe the step-by-step *safe* procedure for replacing a smartphone battery.
4. Explain the difference between a DIMM and a SO-DIMM, and where you would find each.
5. What is a "daughterboard" in a GSM phone, and what components does it typically hold?

WEEK 2: ADVANCED DIAGNOSTIC TOOLS

TOPIC: OSCILLOSCOPE, DC POWER SUPPLY, SOFTWARE TOOLS

Introduction

Basic repairs can be done with a multimeter and screwdrivers. Advanced, component-level repairs require advanced tools. This week, we learn to use tools that *show* us what is happening with electricity (Oscilloscope), *provide* stable power (DC Power Supply), and *talk* to the device's low-level software.

A. The oscilloscope

1. **WHAT IT IS:** A diagnostic tool that displays an electrical signal as a graph of voltage (Y-axis) versus time (X-axis). A multimeter just shows a number; an oscilloscope shows the *shape* of the signal.
2. **KEY USES IN REPAIR:**
 - **Clock Signals:** Checking for clock signals from crystals (e.g., RTC clock, 32kHz). If the clock isn't "ticking," the device won't start.
 - **Data Lines:** Observing data transfer on buses like I2C or SPI (e.g., between the CPU and a sensor).
 - **Power Rail Ripple:** A PSU or PMIC should provide *clean*, flat DC voltage. The oscilloscope can show "ripple" or noise on a power rail, indicating a failing capacitor.
3. **BASIC CONTROLS:**
 - **Volts/Div (Voltage per Division):** Sets the scale of the Y-axis (height).
 - **Time/Div (Time per Division):** Sets the scale of the X-axis (width/speed).
 - **Trigger:** Stabilizes a repeating waveform on the screen.

B. The DC power supply (bench supply)

1. **WHAT IT IS:** A device that provides a clean, stable, and *adjustable* Direct Current (DC) voltage and current.

2. KEY USES IN REPAIR:

- **Powering a Device Without a Battery:** Essential for phones. You can set it to 4.2V and 2A to simulate a battery and see if the phone powers on.
- **Monitoring Current Draw:** This is the most important use.
 - **No Current Draw:** The power button or PMIC circuit is dead.
 - **High Current Draw (Short Circuit):** A component is shorted *before* the power button is pressed.
 - **Current Draw on Button Press, Then Drops:** A secondary power rail is shorted, or the CPU/RAM is faulty.
- **Short Circuit Detection:** You can inject a low voltage (e.g., 1V) and high current into a shorted rail. The shorted component will get hot (use thermal camera or rosin flux to find it).

C. Advanced software tools

1. COMPUTER DIAGNOSTICS:

- **BIOS/UEFI Diagnostics:** Built-in hardware tests (for RAM, HDD).
- **MemTest86:** A bootable USB tool for deep, thorough RAM testing.
- **CrystalDiskInfo:** A Windows utility to read the S.M.A.R.T. (Self-Monitoring, Analysis, and Reporting Technology) data from storage drives to predict failure.
- **HWMonitor:** Checks live temperature, voltage, and fan speed sensors.

2. GSM SOFTWARE TOOLS (FLASHING/DIAGNOSTICS):

- **"Flashing" Tools:** Used to rewrite the phone's firmware (Operating System).
 - **Odin:** For Samsung phones.
 - **SP Flash Tool:** For phones with MediaTek (MTK) SoCs.
 - **Qualcomm Flash Image Loader (QFIL):** For phones with Qualcomm SoCs.
- **Use Cases:** Fixing boot loops, removing viruses, upgrading/downgrading firmware.
- **Driver Software:** PC needs specific "drivers" to recognize a phone in flashing mode (e.g., MTK VCOM driver, Qualcomm 9008 driver).

Evaluation questions

1. What is the primary difference between a multimeter and an oscilloscope?
2. Name two uses for an oscilloscope in board-level repair.
3. What is the *most important* diagnostic information you get from a DC power supply when testing a phone?
4. What does a "high current draw" *before* pressing the power button indicate?
5. What is "Odin," and what brand of phone is it used for?

Assignment questions

1. A phone is dead. You connect it to a DC power supply set to 4.2V. Describe what you would expect to see (in terms of current) for the following scenarios: a. A normal, working phone (off). b. A phone with a short circuit on the main power rail. c. A phone with a working power button, but faulty RAM.
2. What is S.M.A.R.T. data, and what software tool can you use to read it?
3. Why would a technician use an oscilloscope to check a power rail instead of a multimeter?
4. What is "firmware flashing," and list two reasons you might need to do it.
5. Explain the function of the "Time/Div" and "Volts/Div" controls on an oscilloscope.

WEEK 3: GSM SCHEMATIC DIAGRAMS

TOPIC: READING AND INTERPRETATION

Introduction

A schematic diagram is the "map" of the logic board. It shows every component and every connection. You cannot perform advanced, component-level repairs without knowing how to read one. It is the single most important skill for a micro-soldering technician.

A. What is a schematic?

- It is **NOT** a picture of the board. It is a logical and electrical diagram.
- It is paired with a **Board Layout (or "Board View")**, which *is* a picture/diagram of the board.
- **How they work together:**
 1. You find a component on the **Board Layout** (e.g., U1001).
 2. You search for "U1001" in the **Schematic** file.
 3. The schematic shows you what U1001 is (e.g., a PMIC) and what every single pin is connected to.

B. Common symbols and designators

- **R:** Resistor
- **C:** Capacitor (C = non-polarized, C or CP = polarized/tantalum)
- **L:** Inductor (coil)
- **D or Z:** Diode or Zener Diode
- **Q:** Transistor (MOSFET, BJT)
- **U or IC:** Integrated Circuit (Chip)
- **J:** Connector (e.g., J1001 = battery connector)
- **X or Y:** Crystal Oscillator
- **TP:** Test Point
- **GND:** Ground (Earth)

- **VCC / VDD:** Positive Power Supply

C. Reading and interpreting

1. SIGNAL NAMES / NETS:

- This is the most important concept. A line on the schematic is a "net" or "signal." It has a name (e.g., PP_VCC_MAIN, LCD_SPI_DATA).
- *Anywhere* you see that same name on the schematic, those points are electrically connected, even if there is no line drawn between them.

2. TRACING A CIRCUIT (EXAMPLE: USB Charging):

- **Start:** Find the USB connector (e.g., J2001) on the schematic.
- **Find VBUS:** One pin will be labeled USB_VBUS_5V or similar. This is the 5V from the charger.
- **Follow the path:** The USB_VBUS_5V line will go through:
 - An Overvoltage Protection (OVP) IC (e.g., U2002).
 - A charging IC (e.g., U2003, "Tigris," "Tristar").
- **Output:** The charging IC will output a new power line (e.g., PP_BATT_VCC) that goes to the battery connector (J1001).
- **Diagnosis:** If the phone is not charging, you use your multimeter to check for 5V at J2001, then at the *input* of U2002, then the *output* of U2002, etc., until you find where the voltage stops.

3. POWER RAILS:

- Schematics are built around power rails.
- PP_VCC_MAIN or VPH_PWR: The main power rail from the battery/PMIC.
- PP1V8_SDRAM: A 1.8V rail for the RAM.
- PP3V0_NAND: A 3.0V rail for the storage.
- If the phone is dead, you check if all these rails are being created by the PMIC.

Evaluation questions

1. What is the difference between a "schematic" and a "board layout"?
2. What do the following designators stand for: U, R, C, L, J?

3. What is a "net" or "signal name"?
4. If a signal named LCD_RESET_L appears on page 5 and page 12 of a schematic, what does that mean?
5. What is the typical first component a VBUS (charging voltage) line passes through after the connector?

Assignment questions

1. You are troubleshooting a phone with no backlight. Using the concept of a schematic, describe the path you would trace, starting from the LCD connector. (Hint: Look for a net named LCD_BL_ANODE).
2. Draw the schematic symbols for a resistor, a non-polarized capacitor, a diode, and a ground.
3. What is the role of a "Test Point" (TP) on a logic board?
4. Explain how you would use a board layout and a schematic *together* to find a component.
5. What is a "power rail"? Find and list 3 common power rail names you might see on a phone schematic.

WEEK 4: MICRO SOLDERING TECHNIQUES

TOPIC: IC REPLACEMENT, JUMPER TECHNIQUES

Introduction

Micro soldering is the most advanced hardware repair skill. It involves working with components that are often smaller than a grain of rice, right up to large, complex Ball Grid Array (BGA) chips. This requires a steady hand, a good microscope, and precise control of heat.

A. Tools and setup

- **Microscope:** A stereo (binocular) or trinocular microscope is **essential**. You cannot do this by eye.
- **Soldering Iron:** A temperature-controlled station with a fine "micro-pencil" or "J-tip."
- **Hot Air Rework Station:** Used to remove and reflow components. Must have adjustable temperature and airflow.
- **Solder:** Leaded (63/37) solder paste or wire (melts at a lower temp, easier to work with).
- **Flux:** High-quality "no-clean" tacky flux. **This is the most important consumable.** Flux cleans the pads and allows solder to flow correctly.
- **Solder Wick (Desoldering Braid):** To clean pads.
- **Tweezers:** Fine, high-quality, curved tweezers.

B. IC replacement

1. **QFP (Quad Flat Pack) / QFN (Quad Flat No-Lead):**
 - These are chips with "legs" (QFP) or pads (QFN) on the *outside*.
 - **Removal:** Apply flux. Use hot air (e.g., 350°C, medium air) in a circular motion. Gently nudge with tweezers until it moves freely, then lift.
 - **Pad Prep:** Clean the pads on the board with solder wick and flux.
 - **Replacement:** Place the new chip (minding pin 1 orientation). Apply flux. Use hot air to reflow. The chip will "self-center" due to solder's surface tension.
2. **BGA (Ball Grid Array):**

- These chips (like CPUs, PMICs) have hundreds of solder *balls* underneath.
- **Removal:** Apply flux around the chip. Use hot air (e.g., 380°C, low-medium air). Gently nudge until it moves.
- **Pad Prep (Board):** This is the hardest part. The pads must be perfectly clean and flat. Use solder wick to remove all old solder.
- **Reballing (The IC):**
 - You cannot re-install a used BGA chip. You must "reball" it (or use a new, pre-balled chip).
 - This involves placing a "stencil" (a thin metal sheet with holes) over the chip, applying solder paste, and heating it to form new, perfect balls.
- **Replacement:** Align the chip *perfectly* (using alignment dots). Apply hot air (no nudging!) until you see the chip "dance" or "settle" as the balls melt.

C. Jumper techniques

1. **WHAT IT IS:** Used to repair a broken "trace" (the copper "wire" on the board) or a "lifted pad" (where the copper pad was ripped off the board).
2. **THE PROCESS:**
 - **Expose Copper:** Gently scrape the green "solder mask" off the trace on both sides of the break with a sharp scalpel.
 - **Tin the Pads:** Apply flux and a tiny dab of solder to the exposed copper.
 - **The Jumper:** Use a *very* thin "jumper wire" (often Kynar wire or enameled copper wire from a coil).
 - **Solder:** Solder one end of the wire to the first pad, run the wire across the break, and solder it to the second pad. Cut the excess.
 - **Cure:** Apply a drop of "UV solder mask" (a green, curable glue) over the wire to insulate it and hold it in place. Cure it with a UV light.

Evaluation questions

1. What are the three essential pieces of *equipment* for micro soldering?
2. What is "flux," and why is it so important?

3. What is the main difference between a QFP chip and a BGA chip?
4. What is "reballing"?
5. What is a "jumper wire" used for?

Assignment questions

1. Describe the full process for removing and replacing a BGA chip.
2. What is "solder wick," and what is its purpose?
3. A technician accidentally rips a pad off the motherboard while removing a charging port.
Describe, step-by-step, how they would use a jumper wire to fix this.
4. Why is a microscope considered essential for this type of work?
5. What is "UV solder mask," and when would you use it?



WEEK 5: MIDTERM EXAMINATION

This week is for the midterm test, which will cover all content from Weeks 1-4.

- **Topics:** SS2 Review, Advanced Tools (Oscilloscope, DC Supply), Schematics, and Micro Soldering.
- **Format:** A combination of objective questions, short-answer questions ("What is this symbol?"), and scenario-based questions ("How would you diagnose...").



WEEK 6: ADVANCED COMPUTER NETWORKING

TOPIC: IP ADDRESSING, CABLING, WIRELESS SETUP

Introduction

All modern devices are connected. As a hardware technician, you are expected to understand *how* they connect. This week, we cover the fundamentals of the Internet Protocol (IP) that underpins all networking, the physical cables that carry the data, and the wireless signals that have become essential.

A. IP addressing (IPv4)

1. **WHAT IT IS:** An IP address (e.g., 192.168.1.10) is a unique logical address for a device on a network, like a house number.
2. **COMPONENTS:**
 - **Address:** 192.168.1.10
 - **Subnet Mask:** 255.255.255.0 (This defines which part of the address is the "Network" and which part is the "Host").
 - 192.168.1 is the Network ID (the street).
 - 10 is the Host ID (the house number).
 - **Default Gateway:** 192.168.1.1 (This is the router's address—the "exit" from the street to the rest of the world).
3. **PRIVATE VS. PUBLIC IP:**
 - **Private:** Used *inside* your local network (LAN). (e.g., 192.168.x.x, 10.x.x.x). These are not reachable from the internet.
 - **Public:** Your *one* IP address that the rest of the world sees (assigned by your ISP).
4. **DHCP (Dynamic Host Configuration Protocol):**
 - This is a service (usually on your router) that *automatically* gives an IP address to every device that connects. This is why you don't have to set an IP for your phone.
 - **Static IP:** Manually assigning an IP address. Used for servers, printers, and routers.

B. Structured cabling

1. CABLE TYPES:

- **UTP (Unshielded Twisted Pair):** The standard "Ethernet" or "LAN" cable.
- **Categories:** Cat 5e (1 Gbps), Cat 6 (1-10 Gbps).

2. CONNECTORS & TOOLS:

- **RJ45 Connector:** The "plug" at the end of the cable.
- **Crimping Tool:** Used to attach the RJ45 connector to the cable.
- **Cable Tester:** Used to verify all 8 wires are connected correctly.

3. WIRING STANDARDS (T568A & T568B):

- This is the *order* of the 8 coloured wires.
- **T568B:** The most common standard. (Orange/White, Orange, Green/White, Blue, Blue/White, Green, Brown/White, Brown).
- **Straight-Through Cable:** Both ends are T568B. (Connects a PC to a switch/router).
- **Crossover Cable:** One end is T568A, one is T568B. (Connects PC to PC directly. Mostly obsolete as modern devices auto-detect).

C. Wireless (Wi-Fi) setup

1. **ACCESS POINT (AP):** The device that creates the wireless local area network (WLAN). This is usually built into your home router.

2. KEY SETTINGS:

- **SSID (Service Set Identifier):** The "name" of your Wi-Fi network (e.g., "MyHomeWiFi"). You can "hide" the SSID, but this is not a strong security measure.
- **Channel:** The radio frequency used (1-11). If your Wi-Fi is slow, changing to a less-crowded channel (e.g., 1, 6, or 11) can help.

3. WIRELESS SECURITY:

- **WEP:** (Wired Equivalent Privacy). **Obsolete and insecure.** Do not use.
- **WPA (Wi-Fi Protected Access):** Better, but WPA1 is old.
- **WPA2:** The long-time standard. Secure.

- **WPA3:** The newest, most secure standard.
- **MAC Filtering:** A security feature where you manually list the *only* devices (by their MAC address) allowed to connect.

Evaluation questions

1. What are the three main components of an IP configuration?
2. What is the purpose of DHCP?
3. What is the most common wiring standard for Ethernet cables (T568A or T568B)?
4. What tool is used to attach an RJ45 connector?
5. What is the *most secure* Wi-Fi encryption standard available today?

Assignment questions

1. A device has the IP 192.168.10.50 and subnet mask 255.255.255.0. What is the "Network ID" and what is the "Host ID"?
2. Describe the step-by-step process of making a "straight-through" Ethernet cable.
3. What is "MAC Filtering," and why is it used?
4. Your client's Wi-Fi is slow, but their internet plan is fast. They live in a crowded apartment building. What is the *first* setting you would change on their router?
5. What is the difference between a "Private IP" and a "Public IP"?

WEEK 7: MIDTERM BREAK



WEEK 8: REPAIR CASE STUDIES

TOPIC: REAL-LIFE GSM/COMPUTER REPAIR SCENARIOS

Introduction

Theory is one thing, but real-world repair is about logical problem-solving. This week, we will walk through real repair scenarios from symptom to solution, applying all the skills we have learned (diagnostics, tools, schematics, soldering).

A. Case study 1: Computer (No POST)

- **SYMPTOM:** Customer brings in a desktop computer. When the power button is pressed, the fans spin, lights come on, but there is no "beep" and no display on the monitor.
- **DIAGNOSTIC PROCESS:**
 1. **External Checks:** Confirm monitor is on and cable (HDMI/DP) is good. (Done, no change).
 2. **Internal (Minimal Configuration):** This is key. A PC only needs a PSU, Motherboard, CPU, and *one* stick of RAM to POST.
 - **Action:** Disconnect *everything* else: HDD, SSD, all USB headers, front panel audio, and the graphics card (if the CPU has integrated graphics).
 - **Result:** Still no POST.
 3. **RAM:** This is the most common culprit.
 - **Action:** Reseat the single RAM stick. (No change). Try the RAM in a different slot. (No change). Try a *known good* stick of RAM. (No change).
 4. **CPU:**
 - **Action:** Reseat the CPU. Check for bent pins in the socket. (Pins look good, reseated, no change).
 5. **PSU:**
 - **Action:** Test the PSU with a PSU tester. (Tester shows all voltages are good).
 6. **Motherboard (CMOS):**

- **Action:** Clear the CMOS by removing the coin-cell battery or using the jumper. This resets all BIOS settings.
 - **Result: BEEP!** The system POSTs and enters BIOS.
- **CONCLUSION:** The BIOS settings had become corrupted (perhaps from a bad overclock or power outage). Clearing the CMOS fixed the issue.
- **FINAL STEPS:** Reconnect all components one by one, testing after each, to ensure no other part is faulty.

B. Case study 2: GSM (Water Damage)

- **SYMPTOM:** Customer dropped their phone (iPhone 8) in water. It is now completely dead.
- **DIAGNOSTIC PROCESS:**
 1. **DO NOT CHARGE IT!** This is rule #1. Charging a wet phone will destroy it.
 2. **Disassembly:** Immediately and completely disassemble the phone. Note the red "Liquid Contact Indicators" (LCIs) are all activated.
 3. **Board Cleaning:**
 - **Action:** Place the logic board in an ultrasonic cleaner with 99% isopropyl alcohol (IPA). This removes all corrosion and mineral deposits from under the chips.
 - **Action:** After cleaning, "bake" the board at a low temp (e.g., 60°C) to ensure all IPA has evaporated.
 4. **Visual Inspection (Microscope):**
 - **Action:** Inspect the entire board for *remaining* corrosion, especially around connectors and capacitors.
 - **Finding:** Heavy corrosion and "burn" marks are visible on capacitors near the backlight driver IC (U5600).
 5. **Schematic & Diode Mode:**
 - **Action:** Use a multimeter in "diode mode" to check the lines around U5600. Diode mode gives a reading of the "resistance to ground," which should match known-good values.

- **Finding:** A capacitor on the PP_LCM_BL_ANODE (backlight power) line shows 0.001—this is a **dead short to ground**.

6. Short Detection:

- **Action:** Use the DC power supply to inject 1V into that shorted line.
- **Finding:** A small capacitor (C5602) immediately smokes and glows. This was the shorted component.

7. Repair:

- **Action:** Use micro-soldering (hot air, tweezers) to remove the bad capacitor (C5602).
- **Action:** Replace C5602 with a new one from a donor board. R* **Result:** The short is gone.
- **CONCLUSION:** Water caused C5602 to short out, pulling the entire backlight circuit down and preventing the phone from booting.
- **FINAL STEPS:** Re-assemble the phone with a new screen (the original was water-damaged) and new battery. Phone boots and functions normally.

Evaluation questions

1. What is the "minimal configuration" to POST a PC?
2. What is the first thing you should *never* do with a water-damaged phone?
3. What tool is used to properly clean a corroded logic board?
4. In Case Study 1, what was the final fix?
5. What multimeter mode is used to find a short circuit on a board?

Assignment questions

1. A customer's laptop fans are "always on high speed," and it is "very hot." Describe the step-by-step *repair* process you would follow.
2. A smartphone's screen is black, but the customer says they can "hear notifications." What is the *most likely* problem, and what two components would you investigate first?
3. A PC user complains of the "Blue Screen of Death" (BSOD) in Windows. List three *hardware* components that are the most likely cause.

4. Why is "re-seating" components (like RAM) a common and effective diagnostic step?
5. In the water damage case, why was it important to use a schematic?



WEEK 9: PROFESSIONAL ETHICS IN ICT REPAIRS

TOPIC: CUSTOMER SERVICE, SAFETY, DATA PRIVACY

Introduction

Your technical skill will get a device fixed. Your professional ethics and customer service will get you a *loyal customer* and build your reputation. Trust is the most valuable currency in the repair business. This week, we cover the "soft skills" and rules that are just as important as your technical skills.

A. Customer service

1. THE GREETING & CHECK-IN:

- Be polite, professional, and empathetic. Their device is broken, and they are stressed.
- **Listen!** Let the customer describe the problem fully.
- **Ask questions:** "When did this start?" "Was it dropped?" "Any water?"
- **Manage expectations:** Be honest. "A water damage repair is complex; I can't guarantee a fix, but I will do my best."
- **The Job Sheet:** *Always* use one. Record the customer's name, number, device model, serial number, and a *detailed* description of the problem. Get a customer signature.

2. COMMUNICATION:

- **No Jargon:** Do not say, "Your PMIC's LDO rail is shorted." Say, "There is a short circuit on the power chip. I need to replace it."
- **Give updates:** If a repair is taking longer, call the customer *before* they call you.
- **Get permission:** *Never* perform an *additional* repair for more money without calling the customer for approval first. (e.g., "The screen is fixed, but I see the battery is also bad. It will be an extra ₱5000. Do you want to proceed?").

B. Safety

1. PERSONAL SAFETY:

- **Electrical:** Use an isolation transformer for high-voltage work (PSUs, monitors).
- **Fumes:** Use a fume extractor when soldering. Solder flux fumes are toxic.
- **Sharps:** Dispose of broken glass, blades, and needles in a sharps container.
- **Battery:** Li-ion batteries can *explode* if punctured. Be extremely careful when prying them out. Never use metal tools; use plastic spudgers.

2. DEVICE SAFETY (ESD):

- **Electrostatic Discharge (ESD).** A tiny shock from your finger can kill a CPU or IC.
- *Always* use an anti-static wrist strap and an anti-static mat when handling logic boards and RAM.

C. Data privacy (the most important ethic)

1. **THE "GOLDEN RULE":** You are a *repair technician*, not a detective. You have *no business* looking at a customer's files.

2. YOUR RESPONSIBILITY:

- A customer is trusting you with their most private information: photos, bank details, messages.
- **Never** open their "Pictures" folder, "Messages," or "Gallery."
- If you *must* boot the OS to test (e.g., to test the camera), only open the camera app. Do not browse their files.

3. DATA BACKUP & WIPE:

- **Disclaimer:** *Always* warn the customer that a repair (especially software or storage-related) *might* result in data loss. Advise them to back up their device if possible.
- **Factory Reset:** If a repair requires a factory reset (e.g., fixing a boot loop), you *must* get the customer's explicit permission first.

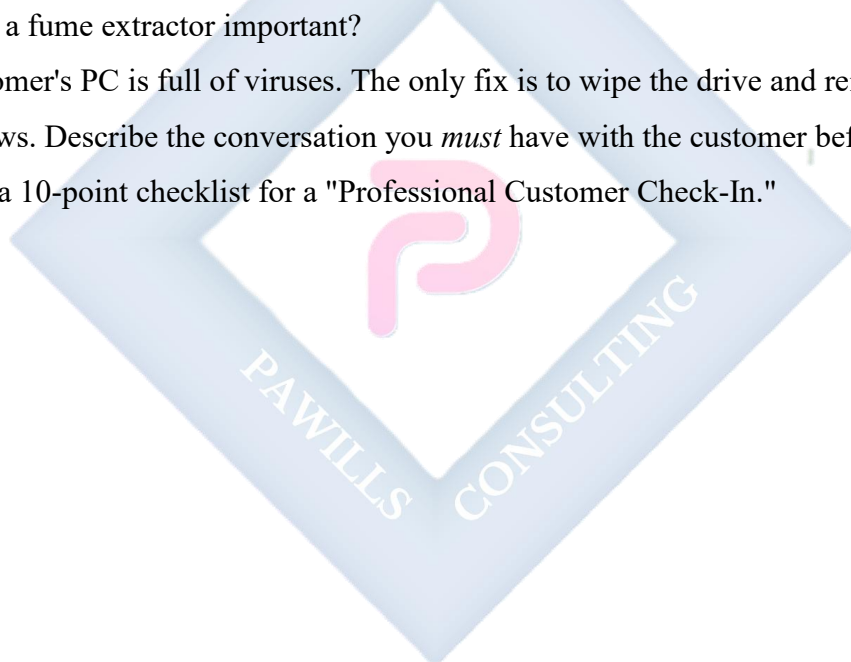
4. **CONSEQUENCES:** Accessing, copying, or sharing a customer's private data is not just unethical—it is a *crime*. It will destroy your reputation and can lead to legal action.

Evaluation questions

1. What is the purpose of a "Job Sheet"?
2. Why is it important to avoid using "jargon" with a customer?
3. What is the biggest personal safety risk when replacing a smartphone battery?
4. What do "ESD" and "anti-static mat" protect?
5. What is the "Golden Rule" of data privacy?

Assignment questions

1. A customer brings in a phone for a screen repair. You fix it. While testing, you see a message from their bank pop up. What do you do?
2. Write a simple, professional script for a phone call to a customer, explaining that their laptop repair will cost *more* than you quoted and why.
3. Why is a fume extractor important?
4. A customer's PC is full of viruses. The only fix is to wipe the drive and reinstall Windows. Describe the conversation you *must* have with the customer before you do this.
5. Create a 10-point checklist for a "Professional Customer Check-In."



WEEK 10: ICT CAREER OPPORTUNITIES

TOPIC: CERTIFICATION PATHWAYS (COMPTIA, CISCO, GSM TRAINING)

Introduction

Having repair skills is a powerful start, but in the formal ICT world, "certifications" prove your knowledge to employers. A certification is a professional standard. This week, we explore the most valuable certifications and career paths you can pursue.

A. Foundational certifications

1. COMPTIA A+:

- **What it is:** The global, vendor-neutral *standard* for "IT Technician."
- **What it covers:** PC hardware, networking, mobile devices, Windows, macOS, Linux, security, and troubleshooting.
- **Career:** This is the #1 certification for any "Help Desk," "IT Support," or "Field Technician" job.

2. COMPTIA NETWORK+:

- **What it is:** The next step up. It proves you understand networking concepts.
- **What it covers:** IP addressing, ports/protocols, routers, switches, network security, and troubleshooting.
- **Career:** Network administrator, Junior Network Engineer.

B. Specialized certifications

1. COMPTIA SECURITY+:

- **What it is:** The global standard for entry-level cybersecurity.
- **What it covers:** Threats, attacks, vulnerabilities, secure network design, and risk management.
- **Career:** Cybersecurity Analyst, Security Administrator.

2. CISCO CCNA (Cisco Certified Network Associate):

- **What it is:** A *vendor-specific* (Cisco) certification. It is much more in-depth than Network+.
- **What it covers:** Configuring, managing, and troubleshooting Cisco routers and switches.
- **Career:** This is a high-value certification for a "Network Engineer" role.

C. GSM and repair-specific training

1. **WHY IT'S DIFFERENT:** GSM (phone) repair is less standardized. There is no single "global" certification.
2. **TRAINING PATHWAYS:**
 - **Specialized Training Centers:** Many private institutes offer 3-6 month "hands-on" courses in advanced GSM repair and micro-soldering.
 - **Online Mentors/Courses:** Many famous repair technicians (e.g., Jessa Jones, iPadRehab) offer high-level online courses in micro-soldering.
 - **Manufacturer Certification:** (e.g., Apple Certified iOS Technician). This allows you to work for an *authorized* service provider. However, they focus on *replacing* parts, not component-level repair.

D. The career paths

- **Path 1: The Entrepreneur (Repair Shop Owner):**
 - **Skills:** GSM repair, PC repair, customer service, business management.
 - **Pros:** Be your own boss, high-profit potential.
 - **Cons:** High risk, long hours.
- **Path 2: The Corporate IT Specialist:**
 - **Skills:** CompTIA A+, Network+.
 - **Role:** Work "in-house" for a company (e.g., a bank, a university) managing their computers, printers, and network.
 - **Pros:** Stable salary, good benefits.
- **Path 3: The Specialized Engineer:**
 - **Skills:** CCNA, Security+, high-level micro-soldering.

- **Role:** Network Engineer (designing networks), Cybersecurity Analyst (protecting data), or a high-level "Tier 4" micro-soldering tech.
- **Pros:** High salary, specialized work.

Evaluation questions

1. What is the "industry standard" certification for a help desk technician?
2. What is the main difference between CompTIA Network+ and Cisco CCNA?
3. Why is the career path for GSM repair different from PC/IT repair?
4. What does a "Network Engineer" do?
5. What certification would you pursue if you wanted a career in cybersecurity?

Assignment questions

1. Research the CompTIA A+ certification. What are the names of the two exams you must pass to get it?
2. Compare "Path 1 (Entrepreneur)" with "Path 2 (Corporate IT)." List two pros and two cons for each.
3. What is a "vendor-neutral" certification, and why is it valuable?
4. If your passion is micro-soldering and board-level repair, what type of *training* would you seek out after leaving school?
5. What is an "authorized service provider," and how is their repair philosophy different from an independent repair shop?

WEEK 11-12: REVISION & EXAMINATION

- **WEEK 11:** Comprehensive review of all topics from Week 1 to Week 10. Focus on the "advanced" topics: oscilloscope/DC supply use, schematic interpretation, micro-soldering steps, and networking.
- **WEEK 12:** Final Term Examination.

SS3 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

SECOND TERM

WEEK 1: GSM ADVANCED REPAIRS

TOPIC: WATER-DAMAGED PHONES, MOTHERBOARD FAULTS

Introduction

This week, we move into the most complex and common advanced repair scenarios: water damage and motherboard faults. A "dead" phone is often not a single broken part but a complex chain of failures. We will build on Term 1's micro-soldering skills, applying them to real-world diagnostics.

A. Water-damaged phone diagnostics

1. **RULE #1: DO NOT CHARGE!** The first action a customer takes is often to plug in a wet phone. This sends 5 volts through wet, corroded pathways, destroying components and traces. Your first instruction to the customer is to *never* do this.
2. **Immediate disassembly:** Time is the enemy. The phone must be completely disassembled, and the logic board removed.
3. **LCI (Liquid Contact Indicator) check:** Locate the small white stickers (LCIs) on the board and chassis. If they are red or pink, it confirms liquid entry.
4. **Visual inspection (Microscope):** Look for obvious corrosion (white or green "fuzz"), burnt components, or discoloured shields. Pay close attention to connectors (LCD, battery, charging) and power management ICs (PMICs).

B. The cleaning process

1. **Manual pre-clean:** Use a soft, anti-static brush and 99% isopropyl alcohol (IPA) to gently scrub away visible corrosion.
2. **Ultrasonic cleaner:** This is the professional standard.

- **How it works:** The machine uses high-frequency sound waves to create microscopic bubbles in a cleaning solution (like 99% IPA). These bubbles "explode" (cavitates) and scrub the board clean, getting under BGA chips and shields where a brush cannot reach.
- **Process:** Place the board in the cleaner for 5-10 minutes.
- 3. **Drying:** After cleaning, the board must be 100% dry. This can be done by "baking" it at a low temperature (e.g., 50-60°C) for 20-30 minutes or by letting it air-dry.

C. Finding shorts and motherboard faults

- After cleaning, the real diagnostics begin.
1. **DC power supply test:** Connect the board to the DC power supply (set to 4.2V).
 - **High current draw (e.g., 2.0A) before pressing power:** This indicates a "main line short" (e.g., VCC_MAIN is shorted to ground).
 - **No current draw:** The power button circuit or PMIC may be dead.
 - **Small current draw (e.g., 0.05A) on power press, then 0:** This indicates a "secondary rail" short (a power line *after* the PMIC).
 2. **Short detection techniques:**
 - **Rosin flux:** Apply rosin flux to the board. When you inject voltage into the shorted line, the shorted component will get hot and melt the rosin, revealing its exact location.
 - **Thermal camera:** A more advanced tool that shows heat on the board in real-time.
 - **Multimeter (Diode Mode):** Use this to check for shorts on specific capacitors or lines by comparing readings to a known-good board or schematic.
 3. **Repair:** Once the shorted component (usually a capacitor) is found, use micro-soldering techniques (hot air, tweezers) to remove and replace it.

Evaluation questions

1. What is the *first* and *most important* rule for a water-damaged phone?
2. What does "LCI" stand for, and what does it tell you?

3. How does an ultrasonic cleaner work?
4. On a DC power supply, what does a high current draw *before* pressing the power button mean?
5. What is the "rosin flux" method used for in short detection?

Assignment questions

1. Describe the full, step-by-step procedure (from customer check-in to re-assembly) for handling a water-damaged phone.
2. A phone board is connected to a DC supply. It draws 0.0A. You press the power button, and it draws 0.08A, then immediately drops back to 0.0A. What type of fault does this *most likely* indicate?
3. What is the difference between a "main line short" and a "secondary rail short"?
4. Why is 99% Isopropyl Alcohol (IPA) used for cleaning electronics instead of water or 70% alcohol?
5. What are the two most common components to fail and cause a short circuit after water damage?

WEEK 2: SMARTPHONE TECHNOLOGY

TOPIC: ANDROID VS IOS ARCHITECTURE

Introduction

As a technician, you must understand that Android and iOS devices are fundamentally different, not just in their software, but in their hardware philosophy. This difference impacts everything from parts availability and repairability to the types of faults you will encounter.

A. Core philosophy and hardware

1. iOS (Apple):

- **Closed ecosystem:** Apple designs the hardware (SoC, board) and the software (iOS).
- **Unified hardware:** All iPhone 12 models use the same parts, board layout, and chips. This makes troubleshooting consistent.
- **Hardware control:** Apple notoriously "pairs" components. This means a new screen, battery, or camera may not function 100% (e.g., no "True Tone," "Battery Health" warning) unless it is an original Apple part, "paired" to the logic board using their proprietary software.

2. Android (Google):

- **Open ecosystem:** Google makes the OS (Android), but many different companies make the hardware (Samsung, Tecno, Xiaomi, etc.).
- **Fragmented hardware:** There are thousands of different Android models. They use different SoCs (Qualcomm, MediaTek, Exynos), different board layouts, and different parts. A Tecno screen will not work on a Samsung.
- **Repairability:** This varies. Parts for popular brands (like Samsung) are common. Schematics and flashing tools are widely available, especially for MediaTek (MTK) and Qualcomm chips.

B. Software and flashing

1. iOS:

- **Flashing:** Done via iTunes (Windows/Mac) or 3uTools (Windows). The phone is put into "Recovery Mode" or "DFU Mode."
- **Security:** Very high. If a user has an iCloud lock, it is *impossible* for a technician to bypass it. You cannot "flash" away the lock.

2. Android:

- **Flashing:** Uses different tools based on the SoC.
 - **Odin:** For Samsung devices.
 - **SP Flash Tool:** For MediaTek (MTK) devices.
 - **QFIL/MiFlash:** For Qualcomm devices.
- **Security:** Varies. A "Factory Reset Protection" (FRP) lock (Google lock) is common after a reset, but it is often bypassable by a skilled technician. Flashing a new ROM (Read-Only Memory) is a common repair for "boot loop" issues.

C. Common repair differences

Feature	iOS (iPhone)	Android
Common Faults	Charging port IC (Tristar), Audio IC, "Touch Disease" (BGA failure).	Charging ports (physical), broken screens, software/boot loops.
Parts Sourcing	Tightly controlled. High-quality "aftermarket" parts are needed.	Varies. Easy for popular models; impossible for obscure ones.
Schematics	Leaked by non-official sources. High quality but hard to find.	Widely available for many models, especially those with MTK/Qualcomm chips.
Component "Pairing"	Yes. (Screen, Battery, Camera, Face ID).	No. Parts are generally interchangeable (with the same model).

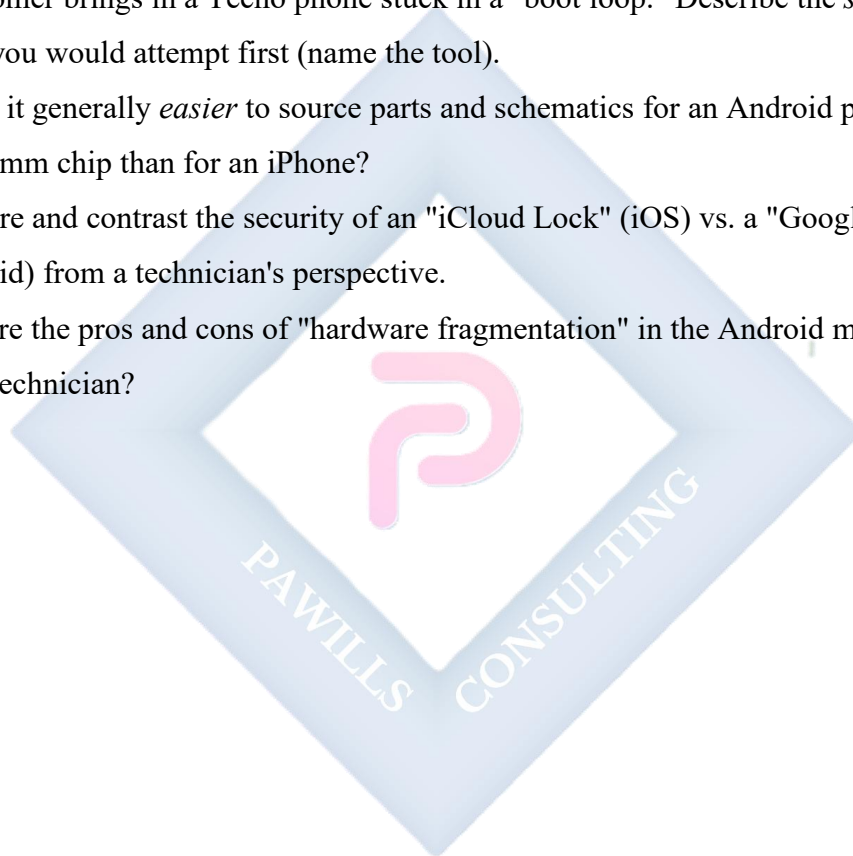
Evaluation questions

1. What is meant by a "closed ecosystem" for iOS?
2. What is "component pairing," and which company is known for it?

3. What software tool is used to flash Samsung phones?
4. What tool is used for MediaTek (MTK) phones?
5. What is "FRP Lock" on an Android phone?

Assignment questions

1. A customer brings you an iPhone with a replaced screen (from another shop) and complains "True Tone is missing." Explain why this is and if you can fix it.
2. A customer brings in a Tecno phone stuck in a "boot loop." Describe the *software-based* repair you would attempt first (name the tool).
3. Why is it generally *easier* to source parts and schematics for an Android phone with a Qualcomm chip than for an iPhone?
4. Compare and contrast the security of an "iCloud Lock" (iOS) vs. a "Google FRP Lock" (Android) from a technician's perspective.
5. What are the pros and cons of "hardware fragmentation" in the Android market for a repair technician?



WEEK 3: ADVANCED COMPUTER MAINTENANCE

TOPIC: SERVER MAINTENANCE, VIRTUALIZATION BASICS

Introduction

While desktop PCs and laptops are common, the backbone of all modern ICT is the "server." Servers are specialized computers that are more reliable, powerful, and complex. As an advanced technician, you may be called to service these systems.

A. What is a server?

A server is a computer designed to run 24/7, 365 days a year, providing "services" to other computers (clients). This requires special hardware.

- **Form factors:**
 - **Tower server:** Looks like a large desktop PC.
 - **Rack-mount server:** A flat, wide chassis (e.g., 1U or 2U) designed to be mounted in a server rack.
- **vs. Desktop:** Servers prioritize reliability and stability, not maximum speed (like a gaming PC).

B. Key server hardware

1. **ECC (Error-Correcting Code) RAM:**
 - Standard RAM (non-ECC) can have a random "bit flip" (a 1 becomes a 0), causing a crash. This is rare and acceptable on a desktop.
 - ECC RAM has an extra chip that can *detect and correct* these errors on the fly. This is essential for a server that cannot crash.
2. **RAID (Redundant Array of Independent Disks):**
 - This uses multiple hard drives to act as one, providing speed and/or data protection.
 - **RAID 0 (Striping):** Combines 2+ disks. Very fast, but *no* redundancy. If one drive fails, *all data is lost*.

- **RAID 1 (Mirroring):** Combines 2 disks. All data is "mirrored" (copied) to both. If one drive fails, the other takes over. This is for *redundancy*.
- **RAID 5 (Striping with Parity):** Uses 3+ disks. It "stripes" data like RAID 0, but also writes "parity" (a check-sum) across all disks. If one drive fails, you can replace it, and the system will "rebuild" the lost data.
- **Hot-swappable drives:** Drives in a RAID array can often be pulled out and replaced *while the server is running*.

3. Redundant power supplies (PSUs):

- Most servers have two (or more) PSUs. If one PSU fails, the other one instantly takes the full load, and the server never powers down.

C. Virtualization basics

1. **WHAT IT IS:** Virtualization is the process of creating a "virtual" version of something, like a computer. It means running multiple, separate "Virtual Machines" (VMs) on *one* physical server.
2. **THE HYPERVISOR:** This is the special software that makes virtualization possible. It sits between the physical hardware and the "guest" operating systems.
 - **Examples:** VMware ESXi, Microsoft Hyper-V, Proxmox.
3. **WHY IT MATTERS TO A TECHNICIAN:**
 - A client might say "Our file server is down."
 - You might go to the server room and see only *one* physical machine. That one machine might be running 10 different "virtual" servers (file server, email server, web server).
 - The problem could be the *physical* hardware (the "host") or the *virtual* software (the "guest"). You must understand this new layer of complexity.

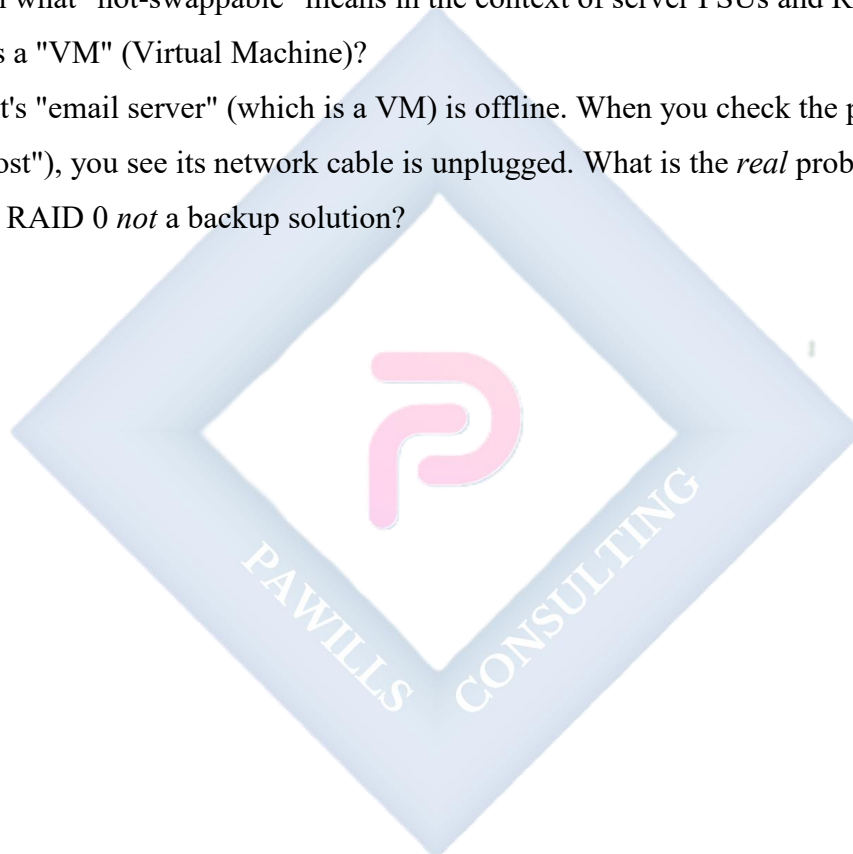
Evaluation questions

1. What is the primary difference in *design philosophy* between a server and a desktop PC?
2. What is ECC RAM, and why is it used in servers?
3. What is the main advantage of RAID 1?

4. What is the main *disadvantage* of RAID 0?
5. What is a "hypervisor"?

Assignment questions

1. A small business has a server with 4 hard drives. They want a balance of *speed* and *redundancy* (ability to survive one drive failure). Which RAID level (0, 1, 5, or 10) would you recommend, and why?
2. Explain what "hot-swappable" means in the context of server PSUs and RAID drives.
3. What is a "VM" (Virtual Machine)?
4. A client's "email server" (which is a VM) is offline. When you check the physical server (the "host"), you see its network cable is unplugged. What is the *real* problem?
5. Why is RAID 0 *not* a backup solution?



WEEK 4: ENTREPRENEURSHIP IN ICT

TOPIC: COSTING, PRICING, BUSINESS PLAN

Introduction

Having technical skills allows you to fix a device. Having business skills allows you to get *paid* for fixing it. This week, we cover the fundamentals of turning your repair skill into a profitable business.

A. Costing: Knowing what you spend

Before you can set a price, you must know your costs.

1. **Fixed costs:** Costs that *do not* change, no matter how many repairs you do.
 - Rent for your shop.
 - Salary (if you have staff, or your own "draw").
 - Tools (this is a one-time cost, but should be "amortized" or spread over time).
 - Utilities (power, internet).
2. **Variable costs (or COGS - Cost of Goods Sold):** Costs that are directly tied to a *single repair*.
 - The replacement part (e.g., the new screen).
 - Consumables (e.g., solder, flux, adhesive).
 - Transportation (if you had to go get the part).

B. Pricing: Setting what you charge

This is the most critical part.

1. **Cost-plus pricing:** The simplest method.
 - $\text{Price} = (\text{Variable Cost} + \text{Labor}) + \text{Profit Margin}$
 - **Labor:** You must pay yourself! Decide on an hourly rate (e.g., ₦5,000/hour). If a repair takes 30 minutes, your labor cost is ₦2,500.
 - **Example:**

- iPhone X Screen Cost (Variable): ₦15,000
- Labor (30 mins @ ₦5k/hr): ₦2,500
- **Total Cost:** ₦17,500
- **Final Price (with ~40% profit):** ₦25,000

2. **Value-based pricing:** Charging a flat fee for a *known* repair. This is what most customers want.

- You calculate your cost-plus price in advance and set a simple menu.
- **Example:** "iPhone X Screen Repair: ₦25,000"
- This is better for the customer because the price is transparent.

C. The simple business plan

A business plan is a "map" for your business. It helps you get funding and guides your decisions.

- **Section 1: Executive summary:** A one-paragraph summary of your whole business. (e.g., "A fast, reliable GSM and PC repair shop for students in Yaba.")
- **Section 2: Services offered:** What will you fix? (e.g., GSM screens, PC formatting, software, unlocking).
- **Section 3: Market analysis:**
 - **Customers:** Who are they? (e.g., students, local businesses).
 - **Competitors:** Who else is doing this? (e.g., "Computer Village"). What makes you *better*? (e.g., "We are more trustworthy," "We offer a 3-month warranty").
- **Section 4: Marketing plan:** How will you get customers? (See Week 8).
- **Section 5: Financial plan:**
 - **Start-up costs:** How much money do you need for tools, rent, and parts?
 - **Pricing:** Your price list (from section B).

Evaluation questions

1. What is the difference between a "fixed cost" and a "variable cost"? Give one example of each.
2. What is "COGS"?
3. What are the two main parts of a "cost-plus" pricing formula?

4. Why is "value-based" (flat-fee) pricing often better for the customer?
5. What is a "market analysis" in a business plan?

Assignment questions

1. You want to offer a "Laptop Formatting & Windows Install" service. This takes you 1.5 hours. You have no "part" costs. If your desired hourly labor rate is ₦4,000, what is a fair price to charge the customer?
2. You are setting up a "Tecno Spark 8 Screen Repair" service.
 - The screen costs you ₦8,000.
 - The adhesive costs ₦200.
 - The repair takes you 1 hour. Your labor rate is ₦3,000/hour.
 - a) What is your *total cost* (COGS + Labor)?
 - b) If you want to make a ₦5,000 *profit* on this job, what is your *final price*?
3. What are "start-up costs"? List 5 start-up costs for a new repair business.
4. Why is it a bad idea to just copy your competitor's prices without knowing your own costs?
5. Write a one-paragraph "Executive Summary" for your future repair business.

WEEK 5: MIDTERM EXAMINATION

This week is for the midterm test, which will cover all content from Weeks 1-4.

- **Topics:** Advanced GSM (water damage, shorts), Android vs. iOS architecture, Server Hardware (RAID, ECC), and Business Planning (Costing, Pricing).



WEEK 6: E-WASTE MANAGEMENT

TOPIC: RECYCLING, SAFE DISPOSAL OF ICT WASTE

Introduction

As a repair technician, you will generate a significant amount of "E-Waste" (Electronic Waste). This waste is not like regular trash; it is highly toxic and hazardous. You have a legal and ethical responsibility to dispose of it safely.

A. What is E-Waste?

E-Waste includes any discarded electronic item:

- Old motherboards, logic boards
- Broken LCD screens
- Dead batteries (Li-ion)
- Old PSUs, cables, and peripherals

B. The hazards: Why it's dangerous

E-Waste contains numerous toxic substances that can harm both people and the environment.

1. **Lead (Pb):** Found in older solder on motherboards and in old CRT (tube) monitors. It is a powerful neurotoxin that damages the brain and nervous system.
2. **Mercury (Hg):** Found in the backlight "fluorescent tubes" (CCFLs) of older LCD monitors and laptops. Also a neurotoxin.
3. **Cadmium (Cd):** Found in older rechargeable batteries (Ni-Cd) and some components. It is a known carcinogen (causes cancer).
4. **BFRs (Brominated Flame Retardants):** Chemicals used in plastic casings (like monitor shells, keyboards) to make them fire-resistant. When burned, they release dioxins, which are highly toxic.
5. **Lithium-ion batteries:** While not as toxic, they are a major *fire and explosion hazard* if punctured or short-circuited.

C. Safe disposal and the 3 R's

1. **REDUCE:** The best solution. By repairing devices, you are extending their life and *reducing* the amount of E-Waste. This is the core of the "Right to Repair" movement.
2. **REUSE:**
 - **Refurbish:** A device that is old but working can be cleaned, formatted, and resold or donated.
 - **Harvesting:** A dead device can be "harvested" for parts. A dead laptop's screen, RAM, and keyboard can be used to fix other laptops. This is a key part of the repair business.
3. **RECYCLE:** This is the final option.
 - **Battery disposal:** Li-ion batteries *must* be separated. They are a fire hazard in regular trash. You must collect them in a safe, fire-proof container and give them to a certified e-waste recycler.
 - **Board disposal:** Logic boards and motherboards should also be collected and sent to a recycler who can chemically extract the valuable (gold, copper) and toxic (lead) materials.
 - **Data destruction:** Before recycling any device with storage (HDD, SSD, phone), you *must* physically destroy the storage or use software to wipe it. This is part of your data privacy duty (Term 1, Week 9).

Evaluation questions

1. What is "E-Waste"?
2. Name three toxic substances found in E-Waste and where you would find them.
3. What is the main *safety hazard* of a lithium-ion battery?
4. What is "harvesting" in the context of E-Waste?
5. What is the "Right to Repair" movement?

Assignment questions

1. Describe the safe and responsible disposal process for a "leaking" or "puffy" (swollen) lithium-ion battery.

2. What is "data destruction," and why is it a critical step before recycling a computer?
3. Explain the 3 R's (Reduce, Reuse, Recycle) as they apply *specifically* to your repair business.
4. A customer gives you their old, dead laptop "for free" when you fix their new one.
Describe the "Reuse" (harvesting) process. What 3-4 components would you immediately test and keep?
5. What are BFRs, and why is *burning* E-Waste a massive environmental and health hazard?



WEEK 7: MIDTERM BREAK



WEEK 8: ICT BUSINESS MARKETING

TOPIC: BRANDING, DIGITAL MARKETING, CUSTOMER RETENTION

Introduction

You can be the best technician in the world, but if no one knows you exist, you will have no business. Marketing is the process of getting and keeping customers.

A. Branding: Your reputation

Your "brand" is not just your logo. It is what people *think* and *feel* about you. In the repair business, your brand must equal **TRUST**.

- **Logo and name:** Choose something clear and professional. "Deji's Fast Repairs" is better than "Ultimate Tech Lordz."
- **Professionalism:** Your shop must be clean. You must be polite (Term 1, Week 9).
- **Warranty:** The best way to build trust is to offer a warranty (e.g., 30-90 days) on your repair. This shows you are confident in your work.

B. Digital marketing: Finding new customers

This is the most cost-effective way to get new clients.

1. Google Business Profile (GBP):

- This is the **most important** marketing tool. It is FREE.
- It puts your business on Google Maps (e.g., "GSM repair near me").
- It allows customers to call you, see your hours, and *leave reviews*.
- **Get reviews!** Politely ask every happy customer to leave a review. 5 stars on Google is your best advertisement.

2. Social media:

- **Instagram / TikTok:** Perfect for a repair business. Post short videos of your work: "Before & After" (dirty inside vs. clean inside), a satisfying "screen peel" on a new screen, a micro-soldering repair (timelapse). This *shows* your skill.

- **Facebook:** Good for building a community page, sharing your Google reviews, and answering questions.
- **WhatsApp:** Use a "Business" account. Your "Status" is a great place to post your "deals of the day" or a successful repair photo.

C. Customer retention: Keeping old customers

It is 5x cheaper to keep an old customer than to find a new one.

1. **Keep a database:** A simple book (or Excel sheet) with Name | Phone Number | Device | Repair.
2. **The follow-up:** One week after the repair, send a simple text: "Hi [Name], this is [Your Name] from [Your Shop]. Just checking to make sure your [Phone] is still working perfectly." This shows you care and builds incredible loyalty.
3. **Loyalty/referral program:** "Bring a friend for a repair, and you both get 10% off."

Evaluation questions

1. In the repair business, your brand must equal what?
2. What is a "warranty," and why is it a good marketing tool?
3. What is the *most important* free digital marketing tool for a local repair shop?
4. Why is Instagram or TikTok a good platform for a repair technician?
5. What is "customer retention"?

Assignment questions

1. Go on Google Maps. Search for "computer repair" in your area. Look at the top 3 results. What do they have in common? (e.g., How many reviews? Do they have photos?).
2. Create a 30-day "Social Media Calendar" for your repair business. List 4 post ideas (one per week).
3. Write a polite, professional SMS template you would send to a customer one week after their repair.
4. Explain the difference between "Marketing" (this week) and "Sales" (Term 1).

5. A customer is angry about a repair that failed (within warranty). How do you handle this to *save* them as a customer?



WEEK 9: FINANCIAL MANAGEMENT

TOPIC: RECORD KEEPING, PROFIT ANALYSIS

Introduction

You are making sales and marketing your business. But are you *actually* making money? If you don't track your finances, you are just "guessing." Financial management is the difference between a "hobby" and a "business."

A. Why keep records?

- To know if you are profitable.
- To see which services make the most money.
- To track your expenses and find areas to save.
- To prepare for taxes or to get a bank loan.

B. Essential records

You don't need complex software. You can start with a simple ledger or exercise book.

1. **Cash flow ledger:** A daily book of all "Money In" and "Money Out."
 - **Money In (Revenue):** All sales from customers.
 - **Money Out (Expenses):** Part Costs (screens, batteries), Shop Costs (rent, power, internet), Personal Costs (your salary).
2. **Job log:** Your customer check-in book (from Term 1) is also a financial record. It links every sale to a specific job.

C. Simple profit analysis (e.g., at the end of the month)

This simple calculation is the most important one in your business.

1. **Total Revenue:** Add up all the "Money In" for the month. (e.g., ₦500,000)
2. **Total COGS (Cost of Goods Sold):** Add up the cost of all the *parts* you used. (e.g., ₦200,000)

3. **Gross Profit:** This is Total Revenue - Total COGS.
 - $\text{₦}500,000 - \text{₦}200,000 = \text{₦}300,000$ (This is your "profit from repairs")
4. **Total Expenses (Overhead):** Add up your *other* costs.
 - Rent: $\text{₦}50,000$
 - Power: $\text{₦}20,000$
 - Internet: $\text{₦}10,000$
 - **Total Expenses:** $\text{₦}80,000$
5. **Net Profit (Your Take-Home Pay):** This is Gross Profit - Total Expenses.
 - $\text{₦}300,000 - \text{₦}80,000 = \text{₦}220,000$
 - This $\text{₦}220,000$ is your *actual* profit.

Evaluation questions

1. What is the main reason for keeping financial records?
2. What is a "cash flow ledger"?
3. What is the formula for "Gross Profit"?
4. What is the difference between "Gross Profit" and "Net Profit"?
5. What is "COGS"?

Assignment questions

1. You run a repair shop for one month.
 - Total Sales (Revenue): $\text{₦}800,000$
 - Total Cost of all Parts: $\text{₦}350,000$
 - Shop Rent: $\text{₦}70,000$
 - Power & Internet: $\text{₦}30,000$
 - Calculate:
 - a) The Gross Profit
 - b) The Total Expenses (Overhead)
 - c) The Net Profit
2. From the calculation in question 1, what could you do to *increase* your Net Profit next month? (List two ideas).

3. Why is it important to separate your *business* money from your *personal* money?
4. Your Job Log shows you did 50 "screen repairs" but only 5 "water damage" repairs.
However, the water damage repairs had a *much higher* profit margin. What does this tell you?
5. What are the pros and cons of using a simple "exercise book" vs. "Excel/Google Sheets" for your financial records?



WEEK 10: PROJECT PLANNING

TOPIC: PREPARING FOR FINAL YEAR PROJECT

Introduction

Your final year project (in Term 3) is the capstone of your SS3 education. It is your chance to combine all your skills—technical, safety, and business—into a single, practical project. This week, we learn how to *plan* that project.

A. Purpose of the project

- To demonstrate mastery of a specific skill.
- To solve a practical problem.
- To practice research, planning, and documentation.
- To create a "portfolio piece" you can show to future employers.

B. Choosing a feasible topic

Your topic must be *achievable* in a few weeks.

- **Bad topic (too big):** "Building a new Android phone from scratch."
- **Bad topic (too simple):** "How to use a screwdriver."
- **Good topic (Hardware/Repair):** "A diagnostic and repair guide for the 'No Backlight' fault on the Tecno Camon 15."
- **Good topic (Hardware/Build):** "Designing and building a low-cost, solar-powered charging station for GSM phones."
- **Good topic (Business):** "A complete business plan and marketing setup for a new PC repair service targeting [Your School]."
- **Good topic (Research):** "An analysis of common failure points in student laptops and a proposal for a school maintenance program."

C. The project report structure

Your final project will be a *written report* and a *practical demonstration*. The report should include:

1. **Chapter 1: Introduction:** What is the problem you are solving? (e.g., "Students' phones are always breaking..."). What are your "Aims and Objectives"?
2. **Chapter 2: Literature review:** (Briefly) What have other people done? (e.g., "A review of common phone repair techniques...").
3. **Chapter 3: Methodology:** How did you do your project?
 - (For a repair project): "I used a multimeter, schematic, DC supply..."
 - (For a business project): "I created a survey, designed a logo, calculated pricing..."
4. **Chapter 4: Results and implementation:** What happened?
 - (For a repair project): "The fault was found on capacitor C205... The phone is now working."
 - (For a business project): "Here is my price list... Here is my marketing poster."
5. **Chapter 5: Conclusion:** What did you learn? What would you do differently?

D. Planning your timeline

You must plan your time.

- **Week 1:** Choose topic, write Chapter 1.
- **Week 2:** Do the main *practical work* (the repair, the business plan).
- **Week 3:** Write up your results (Chapter 4).
- **Week 4:** Write conclusion (Chapter 5) and submit.

Evaluation questions

1. What is the main purpose of the final year project?
2. What does "feasible" mean in the context of a project topic?
3. Why is "Building a new phone" a bad topic?
4. What is the "Methodology" chapter of a project report?
5. What is the "Results" chapter?

Assignment questions

1. Brainstorm and write down *two* good, feasible project topics that *you* are interested in.
2. Choose one of your topics and write a 1-paragraph "Introduction" for it. (What is the problem you are solving?).
3. For the same topic, list the "Aims and Objectives." (e.g., "Aim: To build a PC. Objectives:
1. To source low-cost parts. 2. To assemble the PC. 3. To install Windows...").
4. What is the difference between a project "Aim" and a project "Objective"?
5. What *tools* and *materials* would you need for the project topic you chose?



WEEK 11-13: REVISION & EXAMINATION

- **WEEK 11:** Comprehensive review of all topics from Term 2. Focus on "real-world" applications: diagnosing water damage, the business of repair (costing, marketing, finance), and server-level hardware.
- **WEEK 12:** Final Term Examination.
- **WEEK 13:** Closing.



SS3 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

THIRD TERM

WEEK 1: FINAL YEAR PROJECT SUPERVISION

TOPIC: PROPOSAL WRITING, APPROVAL

Introduction

This week, you will formally begin your final year project. This project is the culmination of all your training. The first step is to create a "Project Proposal," which is a formal, written plan that explains what you are going to do, why you are doing it, and how you are going to do it. It is a "contract" between you and your supervisor.

A. The purpose of a project proposal

- **To force you to think clearly:** Writing down your idea helps you identify problems and refine your topic.
- **To get approval:** Your supervisor needs to know that your project is feasible (possible), relevant, and not too simple or too complex.
- **To set a clear scope:** It defines exactly what you will and *will not* do. This protects you from having to add extra work later.
- **To provide a roadmap:** It becomes the outline for your final report (Chapters 1-3).

B. Key sections of a project proposal

Your proposal must be a short, formal document containing these sections:

1. **Project title:** Must be clear and descriptive.
 - *Bad:* "GSM Project."
 - *Good:* "The Design and Construction of a 5V Solar-Powered USB Charging Station for GSM Phones."

- *Good:* "A Case Study in Diagnosing and Repairing a 'No Power' Fault on an HP EliteBook 840 G3 Laptop."
2. **Introduction / Background of the study:** (1-2 paragraphs)
- What is the problem? Why is it important?
 - *Example:* "Many students in rural areas lack access to constant electricity to charge their mobile devices. This project aims to solve this by creating a low-cost, portable solar charger..."
3. **Aim and objectives:**
- **Aim:** The *one* main goal of your project. (e.g., "The aim is to build a working solar charger.")
 - **Objectives:** The 3-5 specific, *measurable steps* you will take to achieve the aim.
 - (e.g., "1. To research and select a suitable solar panel and charging circuit. 2. To source the components. 3. To assemble and solder the circuit. 4. To test the charger's output voltage and current.")
4. **Scope and limitations (Delimitation):**
- What are the boundaries?
 - *Example:* "This project will focus on building a *single* USB charging port. It will *not* include a battery storage system. The project is limited by the budget of ₦5,000."
5. **Methodology:**
- How will you do the work? What tools and materials will you use?
 - *Example:* "The project will involve soldering, use of a multimeter to test voltage, a hot glue gun for assembly, and a USB power meter for final testing."
6. **Expected outcome:**
- What will you have at the end? (e.g., "A functional solar charger and a full written report.")

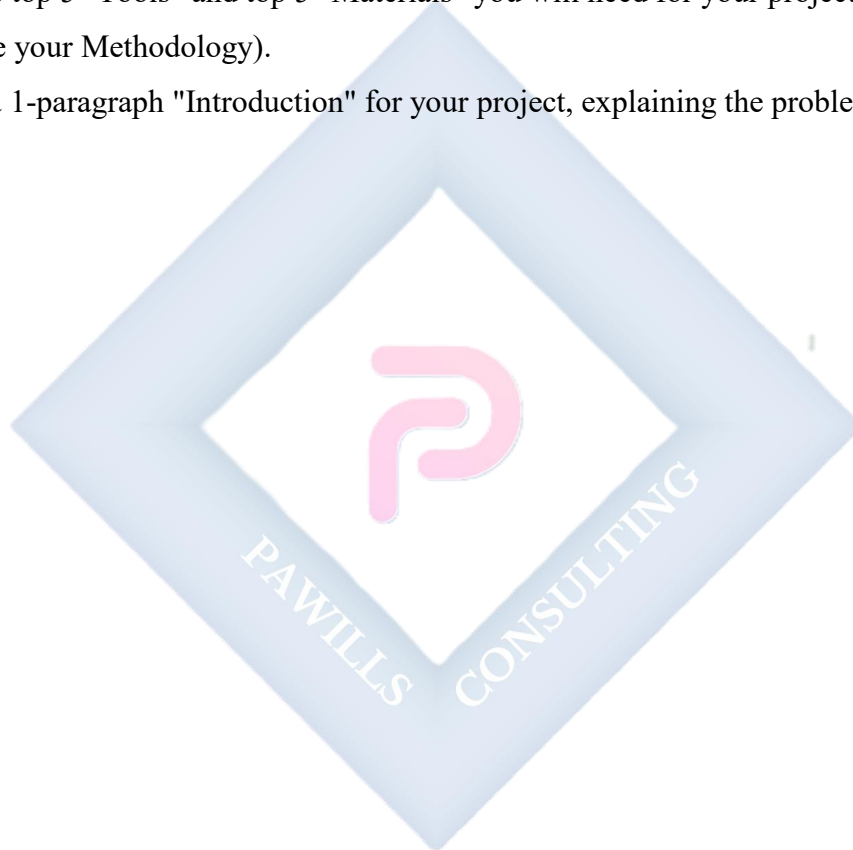
Evaluation questions

1. What is the main purpose of a project proposal?
2. What is the difference between an "Aim" and an "Objective"?
3. Why is the "Scope" section important?

4. What information should be in the "Methodology" section?
5. What is the "Background of the Study" section for?

Assignment questions

1. Write a clear, descriptive "Project Title" for the project you want to do.
2. Write *one* "Aim" and *three* "Objectives" for your chosen project.
3. Write the "Scope" for your project. What will you *not* be doing?
4. List the top 5 "Tools" and top 5 "Materials" you will need for your project (this will become your Methodology).
5. Write a 1-paragraph "Introduction" for your project, explaining the problem it solves.



WEEK 2: PROJECT EXECUTION

TOPIC: HARDWARE OR GSM REPAIR PROJECTS

Introduction

With your proposal approved, the practical work begins now. This phase is all about *doing* the work and, just as importantly, *documenting* every step. Failure to document your work is failure of the project.

A. Safety first: A final reminder

- **ESD (Electrostatic Discharge):** Use your anti-static mat and wrist strap. You can destroy a component without ever seeing a spark.
- **Battery safety:** When opening a phone or laptop, **disconnect the battery first**. A short circuit from a metal tool can cause a fire or destroy the board.
- **Fume safety:** If you are soldering, use a fume extractor or a fan. Solder flux fumes are toxic.
- **Heat safety:** A hot air gun is over 350°C. A soldering iron is 300°C. Be careful.

B. The project logbook (Your "Work Diary")

This is the most critical document you will create. You *must* update it every time you work on your project. It will be used to write your final report.

- Use a dedicated exercise book. Each entry must have:
 - **Date:** e.g., "October 14, 2025"
 - **Task performed:** "Began disassembly of the HP laptop. Disconnected battery, removed motherboard."
 - **Observations / Findings:** "Found a small, dark area of corrosion near the charging port. LCI is red."
 - **Problems encountered:** "One screw was stripped. Had to use pliers to remove it. Broke a plastic stand-off."

- **Next steps:** "Will clean the corroded area with 99% IPA and check the schematic for components in that area."

C. Photographic evidence

You *must* take clear photos at every stage. You cannot take them later.

- **Photo 1: "Before"** (The device, dead, with the problem).
- **Photo 2: "During"** (Close-up of the fault: the corrosion, the broken part, the "no voltage" on the multimeter).
- **Photo 3: "The Fix"** (You soldering the new part, the jumper wire).
- **Photo 4: "After"** (The device working, screen on).

Evaluation questions

1. What is the first component you *must* disconnect when opening a phone or laptop?
2. What does "ESD" stand for, and what two tools help prevent it?
3. What is a "Project Logbook," and why is it important?
4. List the 5 key things to write in a logbook entry.
5. What are the 4 types of photos you must take for your project?

Assignment questions

1. Write your *first* logbook entry for your project (even if it's just "Sourced components" or "Disassembled device").
2. What is the biggest *safety risk* in your specific project? (e.g., "Puncturing the Li-ion battery," "Electric shock from PSU," "Burning myself").
3. Take a clear "Before" photo of your project.
4. What is the *single biggest problem* you are facing right now in your project? (e.g., "I can't find a schematic," "The part is expensive," "I'm not sure where to start").
5. Watch a 10-minute YouTube video related to your project's practical work. Write a 1-paragraph summary of one new skill you learned.

WEEK 3: PROJECT EXECUTION II

TOPIC: PRACTICAL DEMONSTRATION

Introduction

This week is about continuing your work and dealing with the inevitable problems. No project works perfectly the first time. A good technician is just a good troubleshooter. This is also the time to prepare for a "milestone demonstration" to your supervisor.

A. The troubleshooting mindset

When your project doesn't work, do not panic. Follow a logical process.

1. **Check the basics:** Is it plugged in? Is the battery connected? Is the DC power supply on? (You would be surprised how often this is the problem).
2. **What changed?** The fault is almost always related to the *last thing you did*. (e.g., "It worked before I soldered the new chip, so the problem is likely the chip or my soldering.").
3. **Isolate the problem:** Go back to the schematic. Check the voltages. (e.g., "Is the 5V line present? Yes. Is the 3.3V line present? No. Why?").
4. **Review your logbook:** Your diary of what you did is your best guide.
5. **Ask for help (Peer Review):** Get a "fresh set of eyes." Show your project to a classmate. Explain what you've done. Often, just explaining the problem out loud helps you find the solution.

B. Preparing for demonstration

Your supervisor will want to see your progress. You must be able to:

1. **Show your logbook:** It proves you have been working.
2. **Show your practical work:** The device or circuit, even if it's not finished.
3. **Explain your "blockers":** Clearly state what problems you are stuck on. (e.g., "I'm stuck at this point. I've tested this and this, and I don't know what to test next.").

4. **Demonstrate the function:** If it *is* working, be ready to show it.

Evaluation questions

1. What is the "first rule" of troubleshooting when a project fails?
2. What does "Isolate the problem" mean?
3. Why is a "fresh set of eyes" (peer review) a good troubleshooting step?
4. What two items are *most* important to show your supervisor to prove your progress?
5. What is a project "blocker"?

Assignment questions

1. Write a logbook entry for this week, even if it's "Troubleshooting a problem." Detail what you tested and what the result was.
2. What is the *current status* of your project? (e.g., "Disassembled," "Waiting for part," "Working," "Not working").
3. What is the *next* single step you need to take?
4. Explain your project to a family member or friend. Write down one question they asked you. (This is good practice for your defense).
5. Find one "blocker" you have and write down three possible ways to solve it.

WEEK 4: PROJECT REPORT WRITING

TOPIC: DOCUMENTATION, DIAGRAMS, ANALYSIS

Introduction

The project is not finished until the report is written. A perfectly-repaired device with no report is a failure. The report proves you did the work and, more importantly, *you understood the work*. This is the single most important document of your SS3.

A. The 5-chapter project structure (WAEC/NECO Standard)

Your final report must follow this format.

- **Cover page:** Your Title, Your Name, School, Date.
- **Table of contents.**

1. CHAPTER 1: INTRODUCTION

- (This is just an expansion of your proposal).
- Background of the Study
- Problem Statement
- Aim and Objectives
- Scope and Limitations
- Significance of the Project (Why it matters)

2. CHAPTER 2: LITERATURE REVIEW (OR "THEORETICAL FRAMEWORK")

- This is the "theory" behind your project.
- *Example:* If your project is a repair, this chapter explains "The Power-On Sequence of a Laptop," "How to Read Schematics," "Principles of Micro-soldering."
- *Example:* If your project is a solar charger, this chapter explains "How a Solar Panel Works," "How a 5V Regulator IC Works."

3. CHAPTER 3: METHODOLOGY

- This is the "How-To" chapter. It explains *exactly* how you did your work, step-by-step, so someone could (in theory) repeat it.
- List all Tools Used.
- List all Materials Used.
- Describe your *process* (use your logbook!). (e.g., "First, the device was disassembled... Second, visual inspection was performed... Third, the multimeter was used to test...").

4. CHAPTER 4: RESULTS AND ANALYSIS

- **This is the most important chapter.**
- **Results:** What happened? Show your findings. *This is where your photos go!* (e.g., "Figure 1 shows the corrosion found... Figure 2 shows the multimeter reading 0V...").
- **Analysis:** What does it *mean*? (e.g., "The 0V reading confirmed a short circuit. The corrosion on capacitor C5110 was the cause. After replacing C5110, the short was cleared, and voltage was restored, as seen in Figure 3...").

5. CHAPTER 5: CONCLUSION, SUMMARY, RECOMMENDATIONS

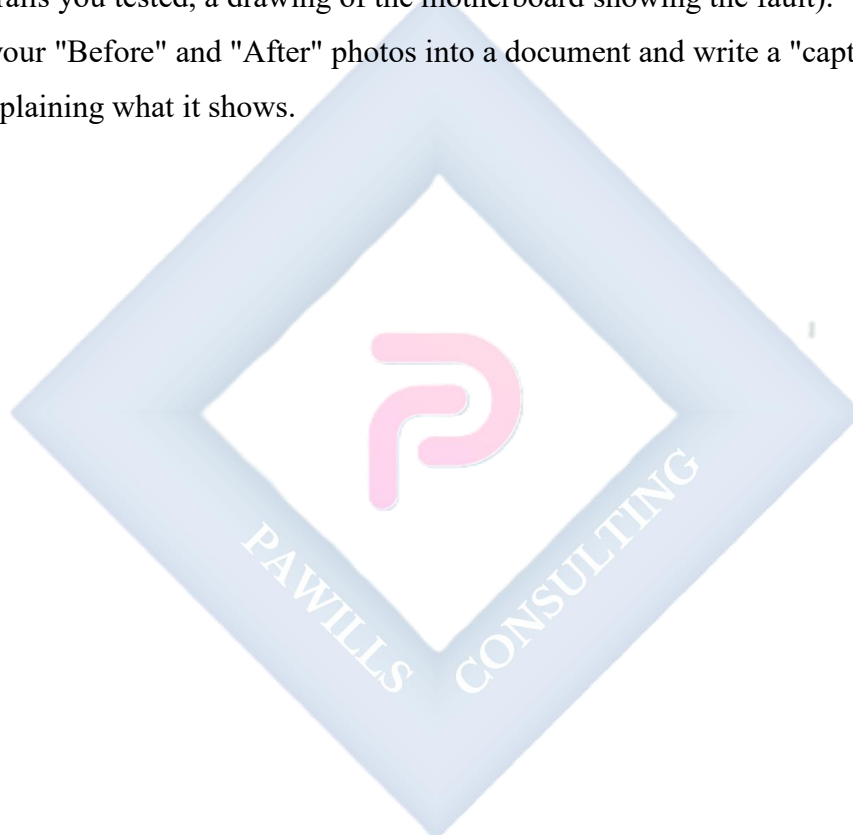
- **Summary:** (1 paragraph) A quick summary of the whole project (Problem, Method, Result).
- **Conclusion:** (1 paragraph) What did you *learn*? Was the project a success? (e.g., "The project was a success and demonstrated that most 'dead' phones can be repaired at the component level...").
- **Recommendations:** (Bulleted list) How could this work be improved? (e.g., "Future projects should use a thermal camera..." or "The school should invest in an ultrasonic cleaner...").

Evaluation questions

1. What are the five mandatory chapters of a final year project?
2. Which chapter is for "Theory"?
3. Which chapter is for "How-To"?
4. What is the *most important* chapter, where you show your photos and findings?
5. What is the difference between "Conclusion" and "Recommendations"?

Assignment questions

1. **PRACTICAL:** Write your full Chapter 1 (you should have most of this from your proposal).
2. **PRACTICAL:** Write your full Chapter 3 (use your logbook and list all your tools and steps).
3. **PRACTICAL:** Write your full Chapter 5 (Conclusion and Recommendations).
4. Draw one key "diagram" for your project (e.g., a simple circuit, a block diagram of the power rails you tested, a drawing of the motherboard showing the fault).
5. Insert your "Before" and "After" photos into a document and write a "caption" for each one, explaining what it shows.



WEEK 5: MIDTERM EXAMINATION

TOPIC: PROJECT SUBMISSION AND MOCK DEFENSE

Content

This is the submission deadline for the first draft of your project report and a "mock defense" (a practice presentation).

- **Part 1: Report submission:** You will submit your full, 5-chapter report for review.
- **Part 2: Practical demonstration:** You will demonstrate that your project (the device, the circuit) is functional.
- **Part 3: Mock defense:** You will give a 5-minute verbal presentation to your teacher and classmates, explaining what you did and what you learned.

Evaluation questions

1. What are the three parts of your midterm examination?
2. What is the purpose of a "mock defense"?
3. What is the most important thing to show in your "practical demonstration"?
4. What chapter of your report details your findings and photos?
5. What chapter details your step-by-step process?

Assignment questions

1. Check your report for spelling and grammar.
2. Practice your 5-minute presentation speech.
3. Prepare your project/device for the demonstration (e.g., make sure it's charged).
4. Anticipate one question your teacher might ask you about your project, and write down the answer.
5. Review the "feedback" from your supervisor on your report and make the necessary corrections.

WEEK 6: PRESENTATION SKILLS

TOPIC: ORAL DEFENSE, USE OF SLIDES

Introduction

Your "oral defense" is the final part of your project. You will stand and present your work. Your goal is to prove to the examiners that *you* did the work and *you* understand the theory behind it.

A. Creating your presentation slides (e.g., PowerPoint)

Keep it simple, visual, and professional. **Do not write long paragraphs of text.**

- **Slide 1: Title Page:** Project Title, Your Name, School.
- **Slide 2: Introduction:** (1-2 sentences) What was the problem?
- **Slide 3: Aim & Objectives:** (List them from your report).
- **Slide 4: Methodology:** (Use *pictures* of your tools and you working, not just text).
- **Slide 5: Results! (The "Hero" Slide):** Show your best "Before" and "After" photos.
This is the most important slide.
- **Slide 6: Demonstration:** (If possible) Show a short video of the project working.
- **Slide 7: Conclusion:** What did you learn?
- **Slide 8: Thank You & Questions:** (e.g., "Thank you for listening. Any questions?").

B. "The 10-20-30 Rule"

A good guide:

- **10** Slides
- **20** Minutes (or less, your defense will be 5-10 mins)
- **30**-point font size (or larger). This forces you to use less text.

C. Speaking and presenting

1. **Do not read the slides!** The slides are for your audience; your *speech* is from your head.
2. **Make eye contact:** Look at your teacher and classmates.

3. **Speak clearly and confidently:** You are the expert on this topic.
4. **Handle questions:**
 - Listen to the full question.
 - Say "Thank you for your question."
 - Answer it clearly.
 - **If you don't know:** Be honest! *Never* lie. Say, "That is an interesting point. I did not investigate that, but I would test it by..." This shows you can think like a technician.

Evaluation questions

1. What is the main purpose of an oral defense?
2. What is the *worst* thing to put on a presentation slide?
3. What is the "10-20-30" rule?
4. What should you do if an examiner asks you a question you don't know the answer to?
5. What is the "hero slide" of your presentation?

Assignment questions

1. **PRACTICAL:** Create your 8-slide presentation.
2. **PRACTICAL:** Practice your 5-minute presentation speech *out loud*.
3. Record your presentation on your phone and watch it. Write down one thing you want to improve.
4. Write down 3 questions you *expect* to be asked.
5. Write down 3 questions you *hope* you are not asked (and then prepare the answers for them).

WEEK 7: MIDTERM BREAK

- Use this break to finalize your project, report, and presentation slides.



WEEK 8: MOCK EXAMS

TOPIC: PAST WAEC/NECO ICT-RELATED PAPERS

Introduction

With the project finished, our focus now shifts 100% to preparing for your final external examinations (e.g., WAEC Data Processing or Computer Studies, which cover hardware). We will use past questions to find your weak areas.

A. Exam structure (Typical)

- **Paper 1: Objective (MCQ):** 40-50 multiple-choice questions (1 hour). Tests a wide range of knowledge.
- **Paper 2: Theory (Essay):** Answer 4 out of 6 questions (2 hours). Requires in-depth answers.
- **Paper 3: Practical:** Hands-on test on a computer (2 hours). (Often focuses on software like Word, Excel, Database, or CorelDraw, but may include basic networking/hardware ID).

B. Mock session: Timed test

- You will be given a full past paper (e.g., WAEC 2022) to complete under exam conditions.
- **Post-Mock Review:** We will grade the paper and analyze the most difficult questions. This is *more important* than the test itself. It shows us what to study.

C. Key "Hot Topics" from past questions

- **Hardware:** "Functions of..." (CPU, RAM, ROM, Motherboard, PSU).
- **Networking:** "Define..." (LAN, WAN, MAN, IP Address, Router, Switch, Topology).
- **Safety/Ethics:** "What is..." (Virus, Anti-Virus, ESD, Data Privacy, E-Waste).
- **GSM (if included):** "Differentiate between Android and iOS," "What is 'flashing'?"
- **Cabling:** "Draw and label the T568B colour code."

Evaluation questions

1. What are the three typical papers in the external ICT exam?
2. What is the purpose of a "post-mock review"?
3. What does "MCQ" stand for?
4. What is the difference between a LAN and a WAN?
5. What is the function of the PSU?

Assignment questions

1. **PRACTICAL:** Complete the 2021 WAEC Data Processing (or relevant) Theory paper at home.
2. **PRACTICAL:** Create flashcards for 20 common ICT acronyms (e.g., CPU, RAM, ROM, IP, LAN, WAN, ESD, ISP, GSM, OS, UI, BIOS, UEFI, SSD, HDD, PSU, LCD, LED, SoC, PMIC).
3. List the T568B colour code from memory.
4. Explain the function of a Router, a Switch, and a Hub.
5. Write a short essay (1 page) on "The Dangers of E-Waste and Proper Disposal Methods."

WEEK 9: REVISION OF PRACTICAL WORKS

TOPIC: HANDS-ON REVIEW

Introduction

This week is a "boot camp" to refresh your core, hands-on skills that could be tested or (more likely) that form the basis of your theory knowledge. We will rotate through 4 stations.

A. Station 1: PC assembly / "Spot the Part"

- **Skill:** Identify all key components and ports on a motherboard (CPU Socket, RAM slot, PCIe x16, SATA, 24-pin power, 8-pin CPU power, I/O ports).
- **Practice:** Safely remove and reinstall a stick of RAM (DIMM).
- **Safety:** Re-demonstrate ESD safety.

B. Station 2: Networking

- **Skill:** Make a CAT 5e "straight-through" cable (T568B standard).
- **Tools:** Use the crimping tool, wire stripper, and RJ45 plug.
- **Test:** Use the LAN cable tester to confirm "Pass" (all 8 lights).

C. Station 3: GSM "Teardown"

- **Skill:** Identify the main components of a smartphone logic board.
- **Practice:** Use a (non-functional) board and a microscope.
- **Identify:** PMIC, SoC, eMMC (storage), SIM slot, RF (Network) area, camera connectors.

D. Station 4: Software diagnostics

- **Skill:** Use common Windows diagnostic commands.
- **Practice:** Open Command Prompt (cmd).
 - ipconfig: Find your IP, Subnet Mask, and Default Gateway.
 - ping google.com: Test internet connectivity.
 - tracert google.com: Trace the path to a server.

- **Skill:** Enter the BIOS/UEFI on a PC (e.g., press F2, F10, or Del on boot).

Evaluation questions

1. What is the colour code for T568B?
2. What is the first step before removing any component from a phone/laptop?
3. What command do you use to find your PC's IP address?
4. What key do you press to enter BIOS on most PCs?
5. What is the difference between a DIMM and a SO-DIMM?

Assignment questions

1. Draw the T568B colour code (8 wires) and label it.
2. A PC turns on, but there is no display. You hear 3 long beeps. What is the *most likely* problem?
3. A LAN cable tester shows lights 1, 2, 4, 5, 6, 7, 8 light up, but light 3 is dark. What does this mean?
4. A customer's phone is "dead." What is the *first* diagnostic tool you would connect it to (from Term 1)?
5. What is the difference between a PMIC and a SoC?

WEEK 10: WAEC/NECO PREPARATORY CLASSES

TOPIC: KEY AREAS LIKELY TO APPEAR

Introduction

This is our final "cram session." We will focus only on the most valuable, most common questions and topics that appear *every year*.

A. Guaranteed topics (Review)

1. "Functions of..."

- **CPU:** The "brain." Executes instructions.
- **RAM:** *Volatile* (temporary) memory for active programs.
- **ROM:** *Non-volatile* (permanent) memory. Holds the BIOS/boot instructions.
- **ALU (Arithmetic Logic Unit):** Part of the CPU. Does math and logic.
- **CU (Control Unit):** Part of the CPU. Directs all operations.

2. Networking devices

- **Hub:** Dumb. Repeats all traffic to all ports.
- **Switch:** Smart. Sends traffic *only* to the intended port.
- **Router:** Connects *different* networks (e.g., your LAN to the Internet).

3. Storage

- **HDD:** Magnetic, spinning platters, cheaper, slower.
- **SSD:** Flash memory (chips), no moving parts, faster, more durable.

4. Safety & Ethics

- **Virus:** Malicious software.
- **Anti-Virus:** Software that detects and removes viruses.
- **Data Privacy:** Protecting customer information (The "Golden Rule").

B. Likely topics (Review)

1. The PC boot sequence (Simplified):

1. Press Power Button.

2. PSU sends power to Motherboard.
3. CPU loads BIOS/UEFI from ROM.
4. POST (Power-On Self-Test) runs (checks RAM, etc.).
5. BIOS finds the boot device (e.g., HDD/SSD).
6. Bootloader (on the drive) loads the Operating System (e.g., Windows).

2. IP addressing:

- **Private IP:** Used *inside* your LAN (e.g., 192.168.x.x).
- **Public IP:** Your *one* address on the internet.

3. GSM Terms:

- **SoC (System on a Chip):** CPU, GPU, and modem all on one chip.
- **Flashing:** Rewriting the phone's firmware (OS).

Evaluation questions

1. What is the difference between RAM and ROM?
2. What is the difference between a Hub and a Switch?
3. What does "POST" stand for?
4. What is the difference between an HDD and an SSD?
5. What is a "Private IP" address?

Assignment questions

1. Write out the PC Boot Sequence step-by-step from memory.
2. Write a short essay comparing and contrasting a Hub, a Switch, and a Router.
3. What is "firmware flashing," and list two software tools used for it.
4. Draw a simple "Star" network topology.
5. Describe 3 safety precautions you *must* take as a repair technician.

WEEK 11-13: REVISION & EXAMINATIONS

- **Week 11: General Revision:** Final Q&A. Reviewing your marked mock exams. Final project report corrections.
- **Week 12: WAEC/NECO Examination:** The external exams begin.
- **Week 13: Closing/Graduation:**

Final Advice:

- Arrive at your exam hall early.
- Read every question (Theory and Objective) *twice* before answering.
- For theory, "List" means list. "Explain" means write full sentences.
- For practicals, stay calm and follow your steps.
- **You have the knowledge and the skills. Be confident. Good luck!**

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